

Project II – Graphical and Numerical Summaries of Data

Part I

The Fortune Global 500, also known as the Global 500, is an annual ranking of the top 500 corporations worldwide as measured by revenue. In total, the Global 500 generated \$31.2 trillion in revenues in 2014. **Fortune Global Data.xlsx** on Canvas provides a list of the 30 companies with the highest revenues (in billions of dollars) in 2014.

- Generate a histogram in StatCrunch, and **estimate** its shape and center
- Which measures of center and variability would be appropriate for this distribution? Why?
- Generate Summary Stats in StatCrunch, what is the true typical value (measure of center)? How close were you with your estimation?

Part II

The National Health and Nutrition Examination Survey (NHANES) is a unique survey that combines interviews and physical examinations. It includes basic demographic information; questions about topics such as diet, physical activity, and prescription medications; as well as the results of a physical examination measuring a variety of variables, including blood pressure and cholesterol levels. The program began in the early 1960s, and the survey currently examines a nationally representative sample of about 5000 persons each year. You will work with the total cholesterol measurements (mg/dL) obtained from participants in the survey in 2009–2010. To examine changes in cholesterol with age, we consider only the 3044 participants between 20 and 50 years of age and have classified them into the three age categories 20s, 30s, and 40s. Please find **Cholesterol Data.xlsx** on Canvas.

- Generate the most appropriate graph for comparison between different age groups. You can generate several graphs and see which one gives you a better representation of your data. *Hint: I'd go with one that starts with B- and ends with -oxplot.*
- Comment on shapes and outliers. How can you spot outliers?
- The ideal range of total cholesterol is below 200 mg/dL. For individuals with elevated cholesterol levels, prescription drugs are often recommended to lower levels. Among the 3044 participants between 20 and 50 years of age, 4 individuals in their 20s, 24 individuals in their 30s, and 117 individuals in their 40s were taking prescription medications to reduce their cholesterol levels. How do you think your comparison of the distribution would be changed if none of the individuals were taking medication? Explain.

Part III

Construct boxplots for the **full IQ** scores (IQF) for the **low lead level group (group 1)** and the **high lead level group (group 3)** in Data Set “**IQ and Lead.xlsx**” on Canvas. The low lead level group consists of children with low levels of lead in their blood, and the high lead level group consists of children with high levels of lead in their blood. What do the boxplots suggest about the effect of exposure to lead?

Submit a pdf with all graph/summary stat outputs and your analysis.